Faculty of Science and Letters

Study of Physics-Informed Neural Networks for Quantum Systems

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I am also deeply grateful to my family for their unwavering support in all my decisions, regardless of the circumstances.

Abstract

Physics-Informed Neural Networks (PINNs) provide an innovative approach to solving differential equations by embedding physical constraints directly into the neural network structure. Unlike traditional numerical methods, PINNs leverage deep learning frameworks while ensuring adherence to governing physical laws. This study explores the application of PINNs in quantum mechanics, specifically for solving the Schrödinger equation in the Infinite Potential Well problem. One of the key challenges in training PINNs effectively is selecting appropriate training data points to optimize convergence. To address this, quasi-random sampling methods, such as Latin Hypercube Sampling (LHS) and Halton sequences, are employed to improve sample distribution and reduce training bias. These sampling techniques ensure a more efficient representation of the solution space, leading to better generalization and accuracy. Building on these advancements, a PINN model is developed to approximate the eigenfunctions and eigenvalues of the Infinite Potential Well problem. The model incorporates multiple physics-based loss functions, including Schrödinger residual loss, eigenvalue loss, normalization loss, and orthogonality loss. Additionally, a hard boundary condition is imposed through a parametric transformation to ensure physically consistent solutions. Results demonstrate that the PINN model successfully identifies the energy eigenvalues and wavefunctions of the Infinite Potential Well with high accuracy. The combination of physics-informed constraints and quasi-random sampling significantly enhances the model's efficiency and reliability. This study provides a foundational framework for extending PINNs to more complex quantum mechanical problems, offering an alternative computational approach for solving differential equations in physics.

1 Introduction

Scientific research relies on theoretical models and experimental validation, often requiring analytical solutions to differential equations that describe systems in physics, biology, chemistry, and economics. However, for many complex systems, obtaining such solutions is infeasible, making computational methods essential. These methods approximate solutions and enable simulations but can be computationally expensive, limiting their efficiency in large-scale scientific studies.

Machine learning offers alternative approaches for solving differential equations and optimizing simulations. In experimental particle physics, it enhances data analysis efficiency and reduces computational costs. Neural networks are increasingly applied in both theoretical and experimental physics, yet they face challenges such as generalization errors, lack of robustness, and limited interpretability.

To address these challenges, more advanced architectures are needed. PINNs [2] integrate governing physical laws, such as differential equations, into learning models, improving accuracy and interpretability. PINNs have been successfully applied in various domains, including fluid dynamics [3, 4, 5], material science [6, 7], and geophysics [8], but their application to quantum systems remains an emerging research direction.

This work serves as an introductory study on applying PINNs to quantum mechanical systems, focusing on the Infinite Well Problem as a representative case. Due to its well-defined analytical solution and fundamental role in quantum mechanics, the Infinite Well Problem provides a controlled setting to assess the effectiveness of PINNs in solving quantum mechanical equations. In addition to evaluating PINN performance, this study explores the impact of quasi-random sampling on model accuracy. The results provide insights into the potential of PINNs for solving quantum mechanical problems and lay the foundation for further research in this direction.

2 Physics-Informed Machine Learning

This section focuses on Physics-Informed Machine Learning, a machine learning approach designed for physical modeling and simulations. The pursuit of physics involves unraveling the fundamental laws governing nature, requiring a close interplay between experimental and theoretical physics. Theoretical models aim to describe physical phenomena with precision, while experimental data serve as a benchmark for validation. Beyond validation, experimental observations often inspire the development of new theoretical frameworks.

Experimental data inherently contain physical information and can be vast, depending on the complexity of the experiment. For instance, collisions at the Large Hadron Collider occur at a rate of approximately one billion events per second, generating nearly one petabyte of data per second [9]. However, experiments are constrained by instrumental limitations such as sensor noise and missing data. Additionally, rare physical events require extensive data analysis to extract meaningful insights. Thus, collecting data alone is insufficient—advanced data processing techniques are necessary for model validation and discovery.

Traditional computational methods struggle with vast datasets and complex real-world challenges, often making analyses computationally expensive or infeasible. Machine learning techniques, particularly neural networks, provide an alternative by leveraging statistical learning and data-driven tools for efficient data analysis and physical modeling.

Significant advancements in deep learning, such as Convolutional Neural Networks [10] and Recurrent Neural Networks [11], have demonstrated remarkable success in scientific research. Applications range from searching for exotic particles in high-energy physics [12] to predicting the sequence specificity of DNA- and RNA-binding proteins [13]. Machine learning techniques are now widely used for data classification, generative modeling, and scientific analysis.

Despite their success, data-driven models suffer from generalization issues and physically inconsistent predictions due to biases and data limitations. Since these models rely solely on provided data, they lack inherent mechanisms to ensure physical correctness. Physics-Informed Machine Learning addresses this by integrating physical laws into machine learning frameworks, imposing constraints that enhance model reliability. This approach enables predictions that align with both empirical observations and theoretical principles, overcoming key limitations of purely data-driven models [1].

Different approaches exist for embedding physics into machine learning models, each focusing on distinct aspects. Three primary methods are observational biases, inductive biases, and learning biases [1]. Observational biases introduce physical knowledge through experimental data, influencing the model's generalization though only as a relatively weak mechanism. Inductive biases involve designing model architectures that inherently enforce problem-specific constraints, such as symmetries, geometric properties, and invariances. Learning biases introduce physical laws into loss functions—particularly useful when solving differential equations—where governing equations, along with boundary and initial conditions, guide the learning process. A prominent example of this approach is PINNs.

2.1 Physics-Informed Neural Networks

PINNs integrate physical laws into neural networks by embedding differential equations into the model structure. The latent function $u(x, t)$ represents the dependent variable of the governing equations, which take the general form:

$$U + N[u] = 0 \tag{1}$$

where x and t denote spatial and temporal coordinates [2]. PINNs leverage automatic differentiation [14] to incorporate these equations into the loss function, along with initial and boundary conditions, ensuring accurate solution learning. Essentially, the neural network output represents a function approximating the differential equation's solution. This process is illustrated in Figure 1, where the neural network predicts a candidate solution that is refined through backpropagation by enforcing the physics of the system.

Although PINNs address several limitations of traditional machine learning, they also present challenges. Hao et al. [15] highlight key difficulties in optimizing PINNs. Different loss components may converge at different rates or even contradict each other, complicating optimization. To address this, various PINN variants incorporate learning rate annealing strategies. Moreover, PINNs often apply simple weighted penalties to

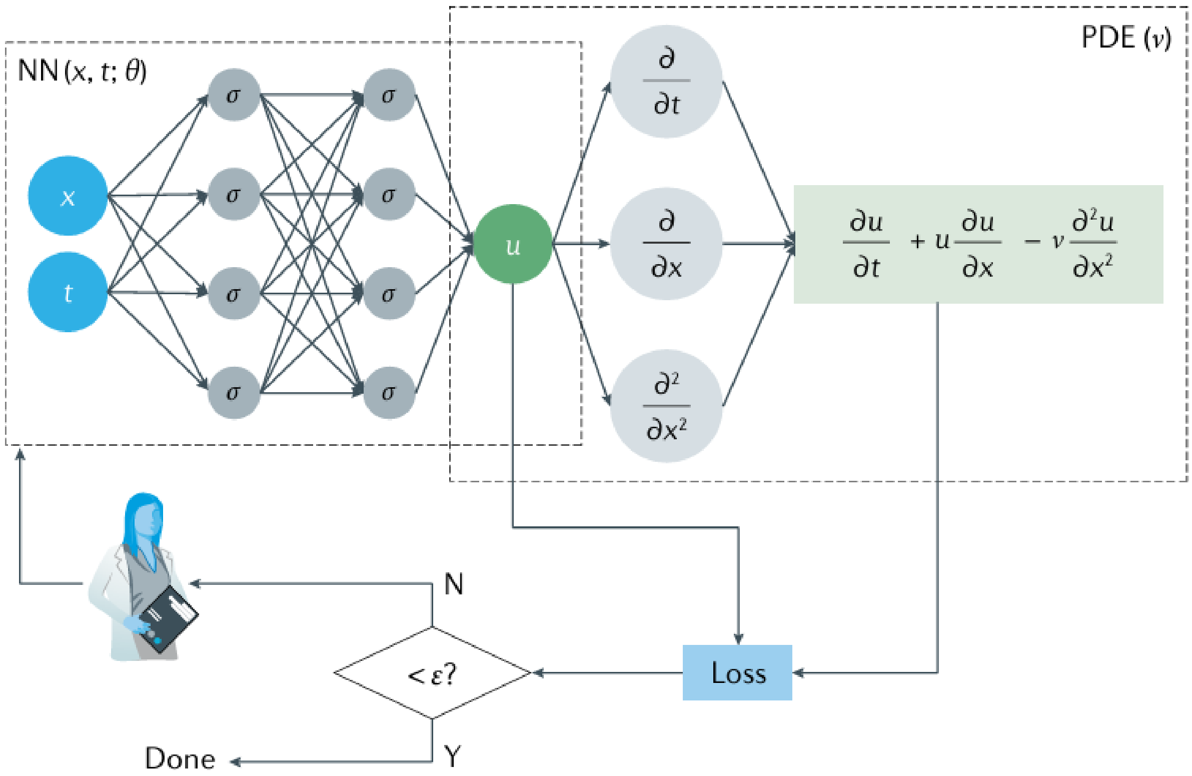


Figure 1: PINN Schematic. [1]

balance PDE residual losses with initial and boundary condition losses, which may be suboptimal for complex equations. Additionally, multi-scale and chaotic physical phenomena, such as shock waves, phase transitions, and turbulence, are difficult to capture with a standard multilayer perceptron architecture.

In particular, quasi-random sampling, a simple yet effective strategy for improving PINN accuracy and efficiency, is investigated.

3 Quasi-Random Sampling

Data plays a crucial role in machine learning methods. Most deep neural network applications rely on data-driven approaches. In this context, data quality is essential, as it should provide general insights and facilitate effective generalization. However, in the case of PINNs, the standard data-driven assumption does not always hold. Since the objective is to find solutions to partial differential equations (PDEs), most applications of PINNs require generating spatial and temporal training data rather than relying on pre-existing datasets. In this context, the Design of Experiment (DoE) becomes a critical factor. DoE ensures that samples are distributed across the entire design space, thereby reducing prediction errors. It includes various techniques for generating training datasets, which serve as the foundation for constructing different surrogate models. Additionally, DoE-based fitting techniques are employed to minimize discrepancies between actual and predicted data.

For an effective exploration of the experimental domain in a computational study, the design must satisfy two key criteria: it should be space-filling to capture as much information as possible from the domain, and it should be non-collapsing, as nearly identical responses provide little new insight and lead to inefficient computational use. In the case of PINNs, to mitigate imbalance in the learning process, the generated spatial and temporal arrays must sufficiently cover the design space. Many sampling methods have been proposed to address this issue, broadly classified into two groups: classical DoE and modern DoE. In this study, quasi-random sampling, a method from modern DoE, is explored.

Quasi-random sampling, also known as low-discrepancy or deterministic sampling, strikes a balance between randomness and regularity to achieve a more uniform distribution of points across a given space. Unlike purely random sampling, which can lead to clustering or gaps, quasi-random sampling follows a structured approach to ensure even coverage, reducing sampling errors and improving representation accuracy. This method is particularly advantageous due to its low discrepancy, which minimizes sampling errors, and its reduced sensitivity to initial conditions. Additionally, it is highly effective in high-dimensional spaces, making it well-suited for numerical simulations and optimization. In machine learning, particularly in PINNs, quasi-random sampling enhances training efficiency by ensuring diverse and well-distributed training points, leading to more stable convergence and better approximations of complex physical systems.

This study focuses on the following quasi-random sampling methods: LHS [16], Halton Sampling [17], and Hammersley Sampling.

3.1 Latin Hypercube Sampling

LHS improves representation accuracy with fewer samples by ensuring that each stratification contains a single randomly selected point. It divides each dimension into N equally spaced intervals, selecting coordinate values randomly or at interval centers while maintaining the Latin property. However, LHS can suffer from poor space-filling properties, especially in high-dimensional spaces. To address this limitation, Maximin Latin Hypercube Sampling (Maximin LHS) improves sample distribution by performing interchanges, swapping coordinate values in selected dimensions to maximize the minimum distance between points. While this enhances coverage, optimizing the process remains computationally expensive due to the large number of possible interchanges.

3.2 Quasi-Random Sequences

The Van der Corput sequence, introduced by J.G. Van der Corput in 1935, generates one-dimensional low-discrepancy sequences over the interval $[0, 1)$. It is considered one of the best-distributed sequences for this range, with no infinitely generated sequence achieving a lower discrepancy. The Halton sequence, proposed by John Halton in 1960, extends the Van der Corput sequence to higher dimensions by generating separate Van der Corput sequences for each dimension using relatively prime bases. The Hammersley sequence, introduced by John Hammersley, an advisor to John Halton, is another variant derived from the Van der Corput sequence and closely resembles the Halton sequence.

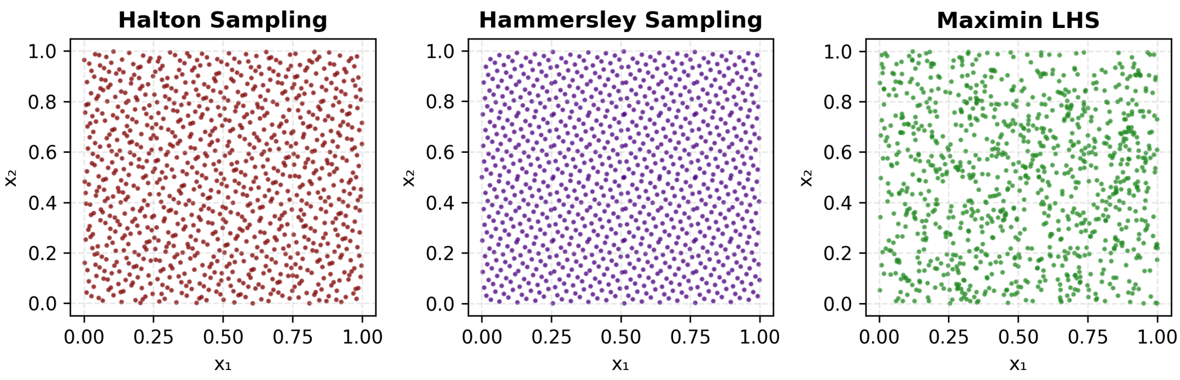


Figure 2: Comparison of quasi-random sampling methods.

Figure 2 compares quasi-random sampling methods with purely random sampling. As expected, quasi-random sampling methods achieve a more uniform distribution across the space compared to purely random sampling. Halton, Hammersley, and Maximin LHS exhibit well-structured, deterministic point distributions that reduce clustering and gaps, improving space-filling properties. In contrast, purely random sampling often results in uneven point clustering, which can lead to suboptimal integration performance.

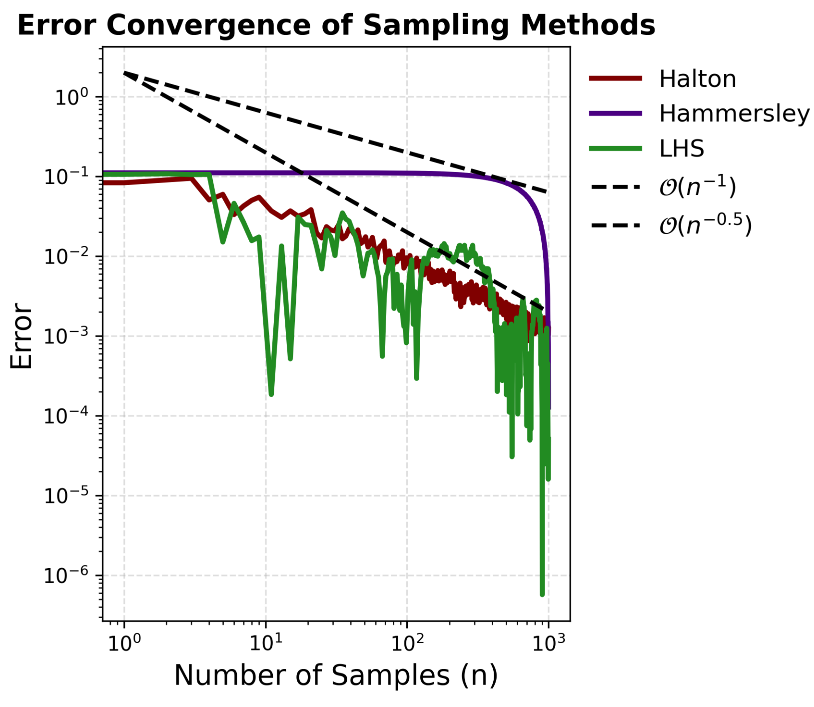


Figure 3: Performance evaluation of sampling methods using Monte Carlo integration.

One approach to evaluating the performance of sampling methods is to compare Monte Carlo integration results with analytical solutions. Figure 3 presents the results of this comparison, where the integration error is plotted as a function of the number of samples used. The error curves for quasi-random sampling methods (Halton, Hammersley, and LHS) show a faster rate of convergence compared to purely random Monte Carlo sampling. The plot includes theoretical error convergence rates of $\mathcal{O}(n^{-1})$ and $\mathcal{O}(n^{-0.5})$, serving as references for expected asymptotic behavior.

The LHS method demonstrates the best performance among the sampling methods, achieving the lowest error for larger sample sizes. Halton and Hammersley sequences, while effective, maintain slightly higher error levels compared to LHS. All three methods significantly outperform random sampling by providing better domain coverage and reducing variance. These results confirm the advantage of using advanced sampling techniques, with LHS emerging as the most accurate approach for numerical integration in this analysis.

4 Design of Quantum PINNs

In this study, PINNs are applied to quantum mechanical systems. The literature contains only a few examples of such applications [18, 19]. This study presents an introductory approach to solving the Infinite Potential Well problem using PINNs, one of the fundamental problems in quantum mechanics. The problem is formulated as an eigenvalue problem:

$$\mathcal{L}f(x) = \lambda f(x) \tag{2}$$

where x represents the spatial variable, $\mathcal{L}$ is a differential operator, $f(x)$ is the eigenfunction, and λ is the corresponding eigenvalue. In quantum mechanics, this equation corresponds to the Schrödinger equation. For the one-dimensional time-independent Infinite Potential Well problem, the Schrödinger equation is given by:

$$\left[\frac{-\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x)\right]\psi = E\psi \tag{3}$$

where the potential is defined as:

$$V(x) = \begin{cases} 0, & 0 \leq x \leq 1 \\ \infty, & \text{otherwise} \end{cases}$$

Setting $\hbar$ and m to 1 for simplicity, the analytical solutions of this eigenvalue problem are:

$$\psi_n(x) = \begin{cases} \sqrt{2}\sin(n\pi x), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E_n = \frac{n^2\pi^2}{2}$$

where n represents the energy states. Therefore, the model is expected to minimize loss at these energy levels and the associated wavefunctions:

$$\psi_1 = \sqrt{2}\sin(\pi x), \qquad E_1 = \frac{\pi^2}{2} = 4.934$$
$$\psi_2 = \sqrt{2}\sin(2\pi x), \qquad E_2 = \frac{4\pi^2}{2} = 19.739$$
$$\psi_3 = \sqrt{2}\sin(3\pi x), \qquad E_3 = \frac{9\pi^2}{2} = 44.413$$

The output of the PINN is a candidate wavefunction corresponding to the Infinite Potential Well problem. The model also predicts an energy value. The goal is to minimize the loss functions for the energy values of the analytical solutions, thereby identifying the wavefunctions in each energy state.

4.1 Loss Functions

A PINN model incorporating physical loss functions is introduced. The total loss function is defined as:

$$\mathcal{L}_{\text{tot}} = \mathcal{L}_{\text{Schrödinger}} + \mathcal{L}_{\text{orthogonality}} + \mathcal{L}_{\text{eigenvalue}} + \mathcal{L}_{\text{normalization}} \tag{4}$$

where each loss component serves a specific purpose:

4.1.1 Schrödinger Loss

The Schrödinger loss ensures that the PINN model satisfies the governing PDE. Since it requires computing derivatives with respect to spatial coordinates, automatic differentiation techniques are used. The loss is formulated as:

$$\mathcal{L}_{\text{Schrödinger}} = \frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\psi + E\psi \tag{5}$$

Minimizing this loss allows the model to determine the correct wavefunction corresponding to the current energy level.

4.1.2 Normalization Loss

In traditional approaches, the linearity of the Schrödinger equation allows the wavefunction to be normalized post-solution without affecting its form. However, when using PINNs, normalization is crucial for maintaining accuracy. A major challenge is that the trivial solution $\psi(x) = 0$ inherently satisfies the equation while minimizing the PDE loss. To prevent this, a specialized loss term enforcing wavefunction normalization is incorporated [19]:

$$\mathcal{L}_{\text{norm}} = \left| \|\psi\|^2 - 1 \right| \tag{6}$$

The integral required for normalization is approximated using Monte Carlo integration, as quasi-random sampling (e.g., Latin Hypercube sampling) is used to generate input data:

$$\mathcal{L}_{\text{norm}} = \left| V(\Omega)\frac{1}{N}\sum_{i=1}^{N}\psi^2(x_i) - 1 \right| \tag{7}$$

However, since this loss function applies normalization as a soft constraint, the wavefunction may still remain unnormalized at the end of training. Therefore, the wavefunction must be explicitly normalized again before obtaining the final result.

4.1.3 Eigenvalue Loss

This is the only non-physical loss in the PINN model. While the Schrödinger loss ensures accurate determination of the first energy state, a mechanism is required to search for higher energy levels:

$$\mathcal{L}_{\text{eigenvalue}} = e^{-E+c} \tag{8}$$

To achieve this, the constant c is incremented by 1 every 5,000 iterations. This adjustment increases the loss, prompting the model to minimize it by exploring higher energy levels. Consequently, this mechanism effectively facilitates the scanning of energy states beyond the ground level.

4.1.4 Orthogonality Loss

To ensure that the network discovers multiple independent eigenfunctions of the Schrödinger equation, an orthogonality loss term is introduced. This is based on the fundamental property of linear differential eigenvalue problems, where solutions are orthogonal:

$$\mathcal{L}_{\text{orth}} = \psi_{\text{eigen}} \cdot \psi \tag{9}$$

where ψ_{eigen} represents the sum of all previously identified eigenfunctions, while ψ corresponds to the network's current predicted solution.

4.2 Parametric Function for Boundary Conditions

Boundary conditions can be enforced using either soft constraints, implemented via loss functions, or hard constraints, directly incorporated into the network's output structure. In this study, a hard boundary condition is applied to ensure that the wavefunction satisfies $\psi(0) = \psi(1) = 0$.

The parametric function used in the model is defined as:

$$\psi(x) = (1 - e^{-(x-0)})(1 - e^{-(1-x)}) \cdot y(x) \tag{10}$$

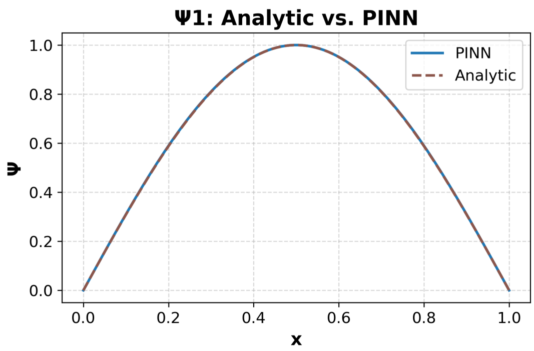
where $y(x)$ represents the neural network's raw output. This transformation ensures the wavefunction automatically satisfies the boundary conditions at $x = 0$ and $x = 1$, removing the need for additional loss terms for boundary constraints.

5 Results

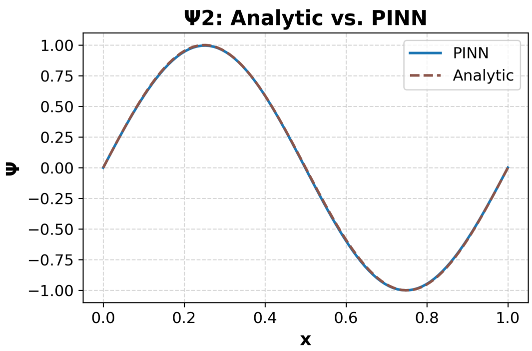
In this chapter, the results of the PINN model trained to solve the Schrödinger equation for the one-dimensional, time-independent Infinite Potential Well problem are presented. The model employs four loss functions: Schrödinger loss, normalization loss, eigenvalue loss, and orthogonality loss, as described in the previous section.

The network architecture consists of an input layer, two hidden layers, and an output layer. The input layer represents the energy level and spatial coordinates, which are merged and passed through the hidden layers. Each hidden layer contains 64 nodes, while the output layer produces a $(64, 1)$ array representing the wavefunction. The sinus function is chosen as the activation function for both the hidden and output layers. Unlike commonly used activation functions such as hyperbolic tangent, the sinus function was selected due to its trigonometric properties, which align with the analytical solution, resulting in better performance.

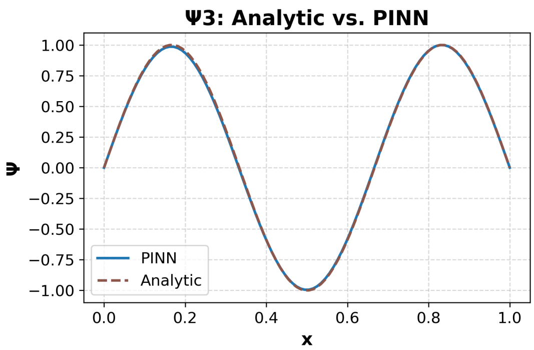
To ensure compliance with the boundary conditions, a hard boundary condition is applied to the output before it is identified as the wavefunction. Additionally, during training, a mechanism is implemented to store the wavefunction based on changes in the loss function. This enables saved wavefunctions to be used in the orthogonality loss term and facilitates retrieving the function after training.



(a) Ψ_1



(b) Ψ_2

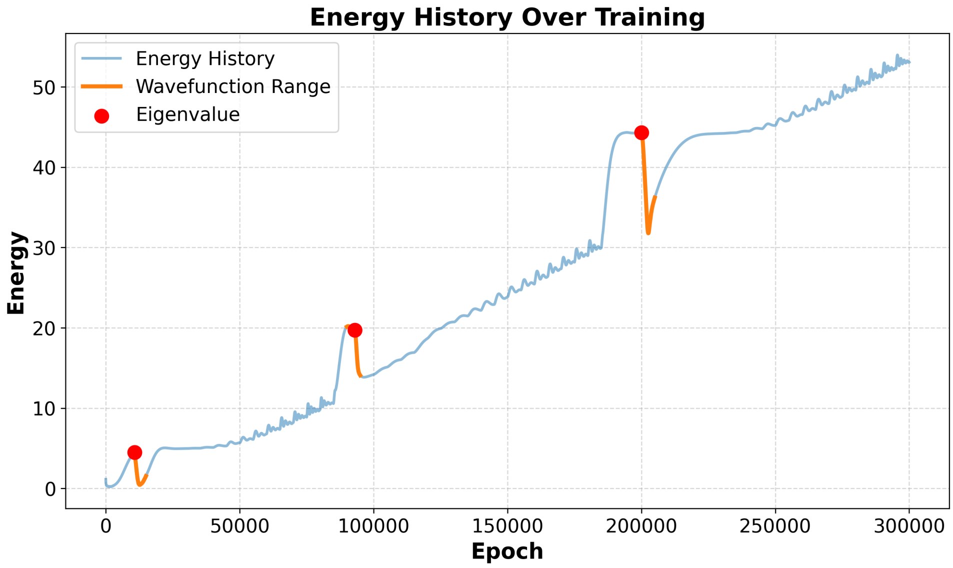


(c) Ψ_3

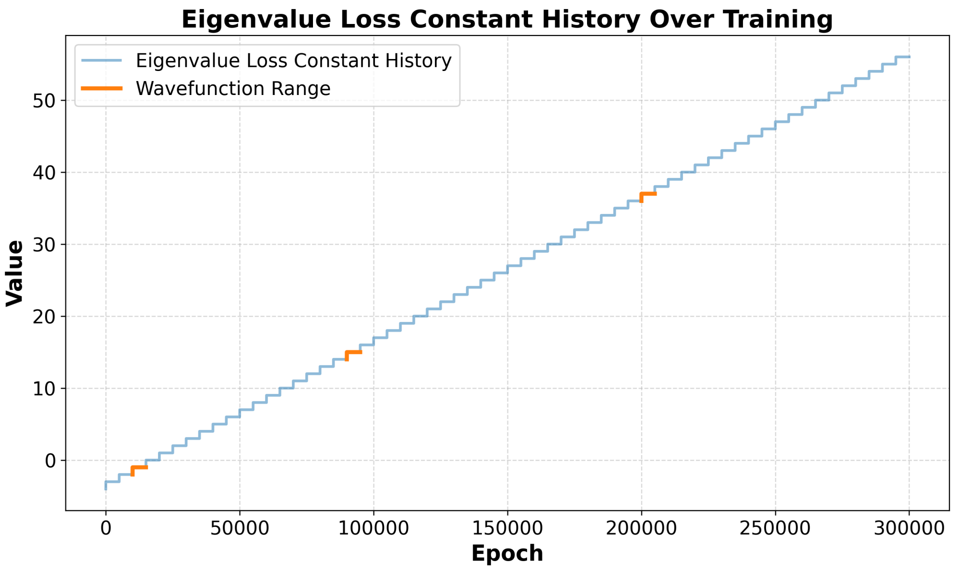
Figure 4: Wavefunctions Ψ_1, Ψ_2, and Ψ_3 obtained from PINN.

The training data is generated using LHS, ensuring well-distributed input points. The model is trained for 300,000 iterations, successfully identifying the first three wavefunctions of the Infinite Potential Well problem.

Figure 4 illustrates the comparison between the PINN solutions and the analytical solutions, demonstrating that the model effectively captures the physics of the problem.



(a) Energy History

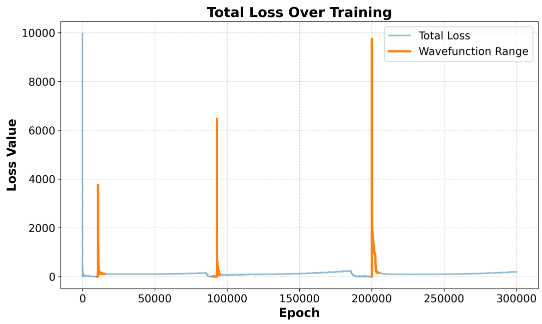


(b) Eigenvalue Loss Constant History

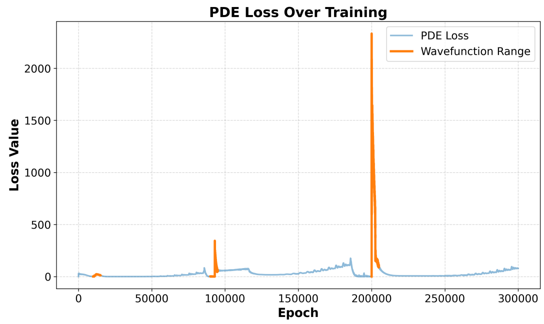
Figure 5: History of energy and eigenvalue loss during training.

Figure 5 presents the energy evolution and the eigenvalue loss constant over the course of training. The eigenvalue loss constant history shows that increasing the constant enables the PINN model to scan higher energy levels. The energy history plot depicts the model's training behavior, where it sequentially scans energy values. Once the model identifies an energy level corresponding to an analytical solution (i.e., it detects a wavefunction), the energy level decreases to a lower state. With the subsequent update to the loss function, the model starts searching for the next wavefunction. This process ensures that each wavefunction is systematically identified, and the plot indicates that finding higher energy wavefunctions requires progressively more iterations.

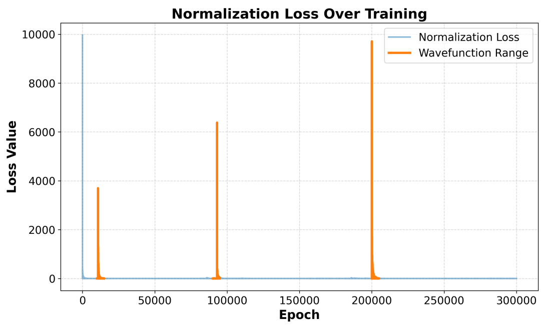
Figure 6 shows the history of the loss functions during training. Similar to the energy evolution, each loss decreases steadily until a wavefunction is found. Afterward, the network is reinitialized, and updates are applied to the loss functions, causing an instant and significant increase in the loss values. This marks the beginning of the process to search for the next wavefunction, during which the loss values decrease again over iterations. These results demonstrate the capability of PINNs to successfully solve fundamental quantum mechanical problems and highlight their potential applications to more complex quantum systems.



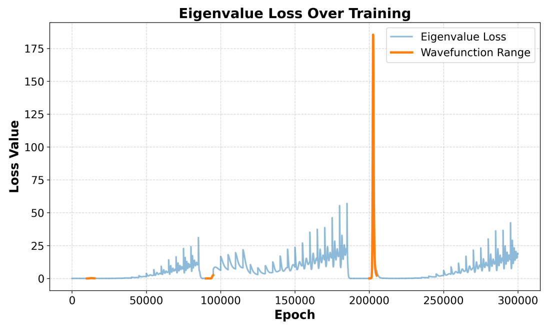
(a) Total Loss



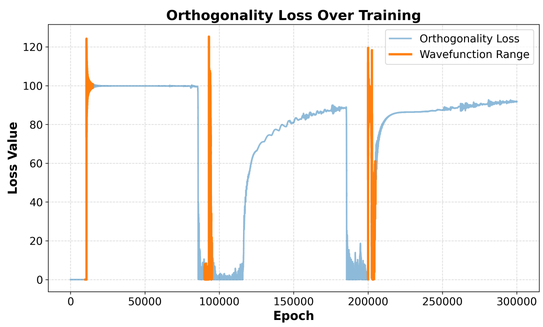
(b) Schrödinger Loss



(c) Normalization Loss



(d) Eigenvalue Loss



(e) Orthogonality Loss

Figure 6: History of loss functions during training.

6 Conclusion

In this study, PINNs were introduced, and quasi-random sampling methods were explored as a strategy to mitigate one of the limitations of PINNs. A PINN model was developed to solve the Infinite Potential Well problem.

The Infinite Potential Well problem is a fundamental topic in quantum mechanics. Solving this problem using PINNs demonstrates their potential to model and learn quantum systems. This study provides an introductory framework for applying PINNs to quantum mechanical problems.

For future work, PINNs can be extended to problems with more complex potential functions and three-dimensional quantum mechanical systems. Additionally, their ability to model quantum systems could be extended to subfields such as particle physics, where PINNs may be integrated into simulations to enhance accuracy and computational efficiency.

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